UNIT 6 · TOPIC 9(B)

Soil Compaction — Effects on Soil and Crop, and Alleviation

Syllabus topic: Soil compaction: causes, methods of alleviation, and effects on soil and crop responses

Effects on the soil

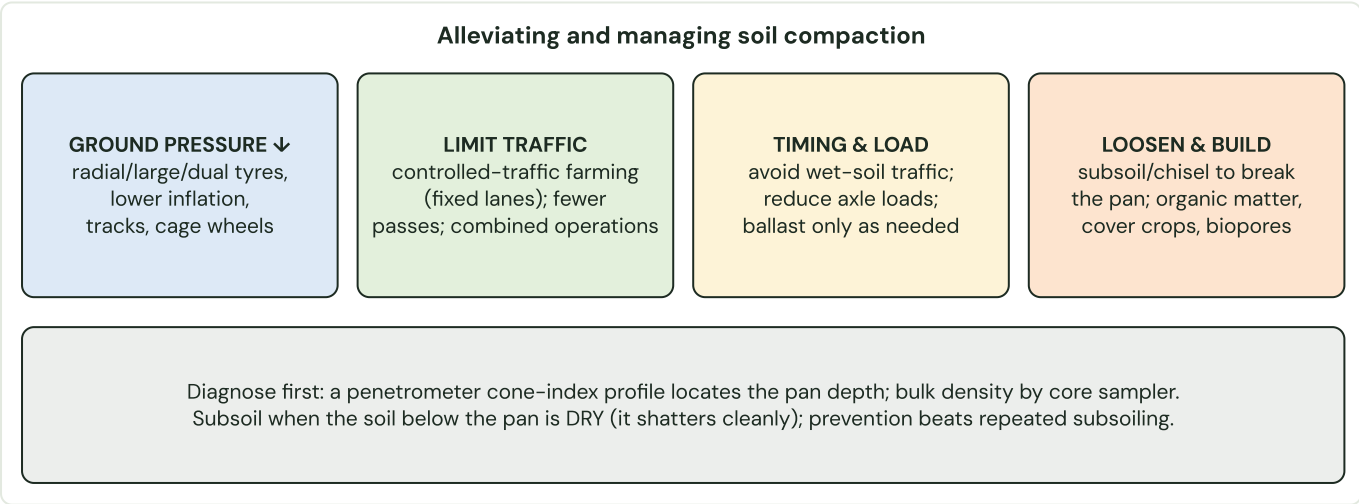
- Bulk density and penetration resistance (cone index) rise.
- Total porosity — and especially the large air-filled MACROPORES — falls, so aeration, infiltration and hydraulic conductivity drop → waterlogging, surface runoff and erosion.
- Soil strength rises, raising the DRAFT and fuel of later tillage (Unit 2, Topic 5).
- A dense sub-surface layer at the usual working depth is the **PLOUGH PAN** (hard pan / tillage pan).

Effects on the crop

- Root growth is mechanically restricted once bulk density / penetration resistance pass a threshold — commonly a root-limiting bulk density of about **1.6–1.8 g/cm³** (depending on texture), or cone index roughly **> 2 MPa**.
- Poor aeration starves roots of O_2; water and nutrient uptake fall.
- Germination and emergence suffer in a crusted or dense seedbed; yields decline.

Moderate surface firming by a press wheel is BENEFICIAL for seed–soil contact — a little compaction around the seed helps, while bulk subsoil compaction harms. Distinguish the two.

Alleviation and management



Alleviating and managing soil compaction (redrawn from the source figure)

PREVENTION — cheapest and most effective:

1. Reduce ground pressure — radial, larger-section or dual tyres, correct/low inflation, tracks or cage wheels.
2. Reduce axle loads; avoid the heaviest loads on wet soil.
3. **CONTROLLED-TRAFFIC FARMING** — confine all wheeling to permanent lanes so the cropped beds stay uncompacted; combine operations to cut the number of passes.

4. Avoid wet-soil traffic (respect the Proctor vulnerability).
5. Build soil resilience — organic matter, cover crops, crop rotation and biopores from deep-rooted crops.

CORRECTION — once a pan exists:

1. Diagnose the pan depth with a cone PENETROMETER profile (Topics 1 and 7) and bulk density with a core sampler.
2. SUBSOIL or CHISEL below the pan to shatter it — most effective when the soil below the pan is DRY and brittle (it shatters cleanly instead of smearing). Winged subsoilers lift and crack a wider zone.
3. Then prevent re-compaction: deep tillage is energy-intensive and only temporary if traffic continues.

Biological loosening (deep-rooted cover crops) and freeze–thaw / wet–dry cycles give slower natural recovery.

Practice MCQs

Q118. Compaction reduces especially the soil's:

(a) Sand fraction
(b) Large air-filled macropores
(c) pH
(d) Colour

Q119. As a result of compaction, water infiltration and hydraulic conductivity:

(a) Increase
(b) Decrease
(c) Stay same
(d) Become zero permanently

Q120. Compaction raises subsequent tillage:

(a) Speed
(b) Draft and fuel use
(c) Quality only
(d) Nothing

Q121. Penetration resistance of compacted soil is measured with a:

(a) Patternator
(b) Cone penetrometer
(c) Hydrometer
(d) Anemometer

Q122. A commonly cited root-limiting bulk density is about:

(a) 0.8 g/cm³
(b) 1.1 g/cm³
(c) 1.6–1.8 g/cm³
(d) 3.0 g/cm³

Q123. A cone index (penetration resistance) that seriously restricts roots is roughly:

(a) >0.2 MPa
(b) >2 MPa
(c) >20 MPa
(d) >200 MPa

Q124. Effects of compaction on the crop include all EXCEPT:

(a) Restricted root growth
(b) Poorer aeration and uptake
(c) Higher yield
(d) Poor emergence

Q125. Firming of soil by a press wheel around the seed is:

(a) Harmful compaction
(b) Beneficial for seed–soil contact and germination
(c) Irrelevant
(d) Illegal

Q126. Which reduces ground pressure and hence compaction?

(a) Narrow, over-inflated tyres
(b) Radial/large/dual tyres and tracks at lower inflation
(c) More axle load
(d) Higher speed

Q127. Controlled-traffic farming works by:

(a) Wheeling the whole field
(b) Confining all traffic to fixed permanent lanes
(c) Random driving
(d) Banning tractors

Q128. Heavy field traffic should especially be avoided when the soil is:

(a) Dry
(b) Wet (near/above field capacity)
(c) Frozen hard
(d) Sandy

Q129. Subsoil compaction from very heavy axle loads is problematic because it is:

(a) Easily reversed by rain
(b) Deep and persistent
(c) Beneficial
(d) Temporary

Q130. A pan is best broken by:

(a) Rolling
(b) Subsoiling/chiselling below the pan
(c) Puddling
(d) More passes

Q131. Subsoiling is most effective (best shattering) when the subsoil is:

(a) Saturated
(b) Dry and brittle
(c) Frozen
(d) Puddled

Q132. To locate the depth of a pan before subsoiling, use a:

(a) Cone-index (penetrometer) profile
(b) Thermometer
(c) pH meter
(d) Tape only

Q133. Building organic matter and using deep-rooted cover crops helps by:

(a) Adding weight
(b) Creating biopores and improving structure/resilience
(c) Raising bulk density
(d) Increasing slip

Q134. Bulk density is measured in the field with a:

(a) Core sampler
(b) Patternator
(c) Bevameter
(d) Tachometer

Q135. The most cost-effective compaction strategy is:

(a) Repeated subsoiling
(b) Prevention (traffic & pressure control)
(c) Deep ploughing every year
(d) Ignoring it

Q139. Compared with prevention, correcting compaction by tillage is:

(a) Cheaper
(b) Energy-intensive and temporary if traffic continues
(c) Permanent
(d) Unnecessary

Q140. Which sequence is correct for managing a compacted field?

(a) Subsoil, then keep wheeling wet
(b) Diagnose pan depth → subsoil dry → then prevent re-compaction
(c) Add ballast and drive more
(d) Do nothing

Answers

Q118 — (b) Large air-filled macropores
Q119 — (b) Decrease
Q120 — (b) Draft and fuel use
Q121 — (b) Cone penetrometer
Q122 — (c) 1.6–1.8 g/cm³
Q123 — (b) >2 MPa
Q124 — (c) Higher yield
Q125 — (b) Beneficial for seed–soil contact and germination

Q126 — (b) Radial/large/dual tyres and tracks at lower inflation

Q127 — (b) Confining all traffic to fixed permanent lanes

Q128 — (b) Wet (near/above field capacity)

Q129 — (b) Deep and persistent

Q130 — (b) Subsoiling/chiselling below the pan — *Subsoiling below the pan shatters best when the subsoil is dry/brittle (smears if wet).*

Q131 — (b) Dry and brittle — *Links to the brittle-vs-plastic stress–strain behaviour of Topic 2.*

Q132 — (a) Cone-index (penetrometer) profile

Q133 — (b) Creating biopores and improving structure/resilience

Q134 — (a) Core sampler

Q135 — (b) Prevention (traffic & pressure control)

Q139 — (b) Energy-intensive and temporary if traffic continues

Q140 — (b) Diagnose pan depth → subsoil dry → then prevent re-compaction

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 10, pp. 253–258 (Table 10-3, first-pass compaction); Gill & Vanden Berg — USDA Agriculture Handbook 316; standard soil-physics texts (Proctor compaction, bulk density); ICAR-CIAE soil-compaction studies; NPTEL soil-dynamics lectures.